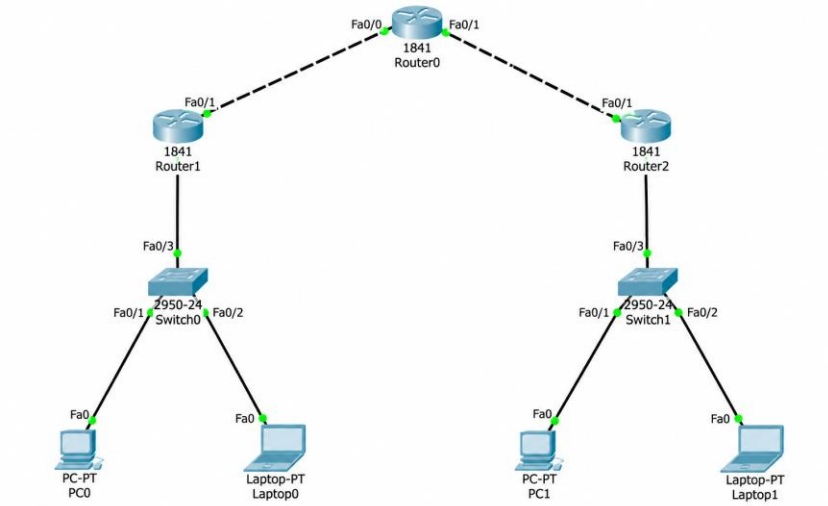
USN : 1WN24CS167

4. How to Configure Static Route and Default Route to the Routers




```
R0>enable
R0#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R0(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1
R0(config)#ip route 40.0.0.0 255.0.0.0 30.0.0.2
R0(config)#exit
R0#
%SYS-5-CONFIG_I: Configured from console by console

R0#write memory
Building configuration...
[OK]
R0#
```

```
R1>enable
R1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 0.0.0.0 0.0.0.0 20.0.0.2
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#write memory
Building configuration...
[OK]
R1#
```

```
R2>enable
R2#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip route 0.0.0.0 0.0.0.0 30.0.0.1
R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#write memory
Building configuration...
[OK]
R2#
```

```
R1>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, l1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 20.0.0.2 to network 0.0.0.0

C    10.0.0.0/8 is directly connected, FastEthernet0/0
C    20.0.0.0/8 is directly connected, FastEthernet0/1
S*   0.0.0.0/0 [1/0] via 20.0.0.2

R1>
```

```
R0>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, l1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 20.0.0.1
C    20.0.0.0/8 is directly connected, FastEthernet0/0
C    30.0.0.0/8 is directly connected, FastEthernet0/1
S    40.0.0.0/8 [1/0] via 30.0.0.2
R0>
```

PC0

Physical | Config | Desktop | Software/Services

Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ping 40.0.0.10

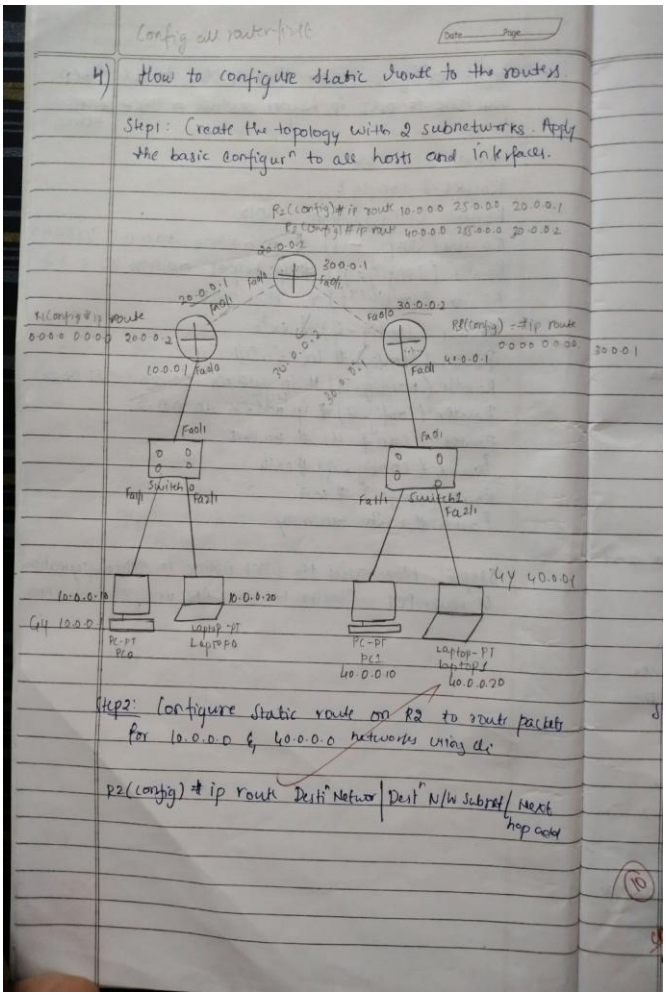
Pinging 40.0.0.10 with 32 bytes of data:

Reply from 40.0.0.10: bytes=32 time=113ms TTL=128
Reply from 40.0.0.10: bytes=32 time=115ms TTL=128
Reply from 40.0.0.10: bytes=32 time=172ms TTL=128
Reply from 40.0.0.10: bytes=32 time=168ms TTL=128

Ping statistics for 40.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 113ms, Maximum = 190ms, Average = 147ms

PC>_
```

Observation Book :



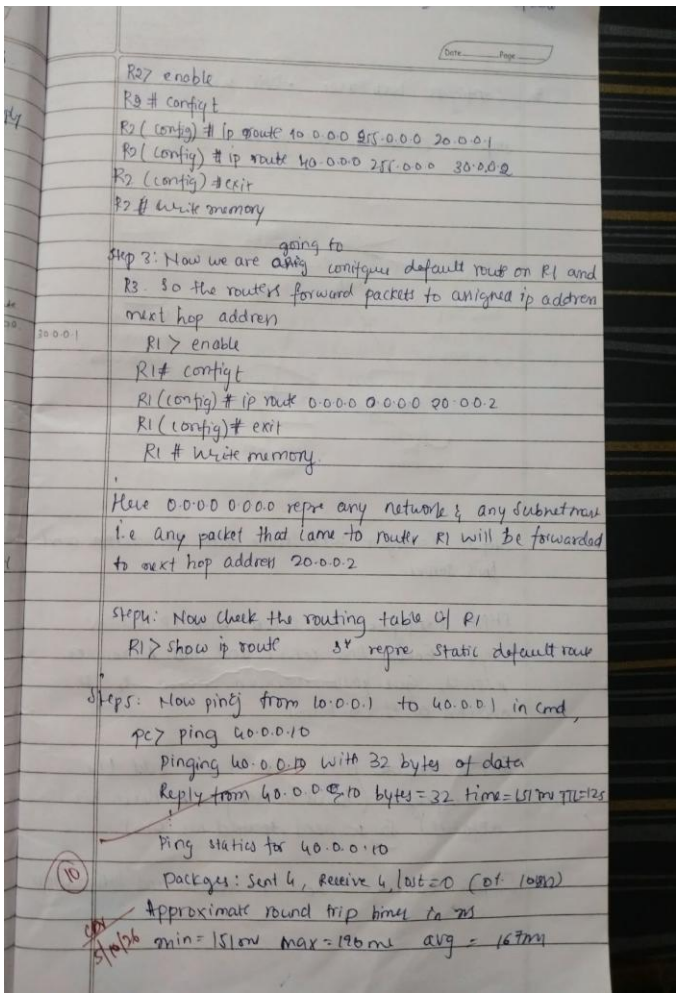
Config all router first

4) How to configure static route to the routers.

Step1: Create the topology with 2 subnetworks. Apply the basic configurⁿ to all hosts and interfaces.

Step2: Configure static route on R2 to route packets for 10.0.0.0 & 40.0.0.0 networks using dᵉ

R2(config)# ip route Destⁿ Networ | Destⁿ N/w Subnet | Next hop addr



R2> enable
R2# configt
R2(config)# ip route 10.0.0.0 255.0.0.0 20.0.0.1
R2(config)# ip route 40.0.0.0 255.0.0.0 30.0.0.2
R2(config)# exit
R2# write memory

Step 3: Now we are going to configure default route on R1 and R3. So the routers forward packets to assigned ip address next hop address

R1> enable
R1# configt
R1(config)# ip route 0.0.0.0 0.0.0.0 20.0.0.2
R1(config)# exit
R1# write memory

Here 0.0.0.0 0.0.0.0 repre any network & any subnet mask i.e any packet that came to router R1 will be forwarded to next hop address 20.0.0.2

Step4: Now check the routing table of R1
R1> show ip route S* repre static default route

Step5: Now ping from 10.0.0.1 to 40.0.0.1 in cmd,
PC> ping 40.0.0.10
Pinging 40.0.0.10 with 32 bytes of data
Reply from 40.0.0.10 bytes=32 time=151ms TTL=125
Ping statics for 40.0.0.10
Packets: Sent 4, Receive 4, lost=0 (0% loss)
Approximate round trip times in ms
Min=151ms max=196ms avg=167ms